1. Clinical Question & Core Hypothesis

Clinical Question

Many MIBC tumors with high intratumoral T cell infiltration and PD-L1 expression nevertheless fail to respond to PD-1/PD-L1 therapy; why does apparent immune engagement not translate to effective antitumor immunity? 2) Tier 1 paths used: (a) PD-L1 ↔ bTMB (core path 1) — suggests neoantigen-driven adaptive PD-L1 upregulation; (b) PD-L1 ↔ CD4/CD8 TILs (core path 3) — links checkpoint expression to tissue T cell presence. I built on these because they directly frame the paradox of immune presence (TIL density) vs. functional response (therapy outcome) and provide a systemic (bTMB) vs. local (PD-L1, TILs) tension. 3) Tier 2 evidence supporting the claim: LRPPRC implicates tumor metabolic/mitochondrial programs that can suppress T cell function; MSH6 (DNA repair) can modulate neoantigen quality/quantity and thus alter the nature of immune priming; SARIFA and systemic metabolic markers (ALP/HDL-C/TG/SHBG/IGF-1) provide microenvironmental and host-state modifiers that could render TILs dysfunctional despite numeric density. 4) Why choose therapy_response as the endpoint: the puzzle concerns failure of immune infiltration to yield clinical benefit under PD-1/PD-L1 blockade, so a therapy-response proxy is the most direct, actionable endpoint for testing mechanism and informing combination strategies. 5) What I deliberately EXCLUDED and why: I limited the core hypothesis to one imaging feature (TIL density) plus two anchor molecular features (PD-L1 and bTMB) to keep the mechanistic question sharp; other Tier 2 modifiers (LRPPRC, MSH6, SARIFA, metabolic panel) are listed as secondary_modifiers for follow-up testing. Tier 3/exploratory paths were excluded from the core to avoid diluting the mechanistic focus?

Working Hypothesis

Primary hypothesis: In muscle-invasive bladder cancer (MIBC), tumors that exhibit high intratumoral T cell density and PD-L1 expression but elevated tumor mutational burden (bTMB) respond poorly to PD-1/PD-L1 blockade due to concomitant TIL dysfunction driven by tumor-intrinsic metabolic programs and systemic host factors. This dysfunction is characterized by T cell exhaustion, impaired cytotoxicity, and an immunosuppressive metabolic microenvironment, which blunt reversal of suppression by checkpoint blockade. Secondary hypotheses: (i) Tumor upregulation of LRPPRC promotes mitochondrial metabolic rewiring that yields a lactate-rich, nutrient-depleted microenvironment, contributing to TIL dysfunction and resistance to PD-1/PD-L1 therapy; (ii) DNA repair context (MSH6 status) shapes neoantigen quality/quantity and presentation, modulating initial T cell priming such that high bTMB with unfavorable neoantigen landscapes correlates with non-response unless metabolic restraining signals are mitigated; (iii) Systemic metabolic host state, as reflected by serum markers (ALP, HDL-C, TG, SHBG, IGF-1) and SARIFA-positivity, correlates with an immunosuppressive milieu that further impairs TIL function and therapy response; (iv) A composite biomarker combining TIL density, PD-L1, bTMB, MSH6 status, tumor LRPPRC expression, and systemic metabolic markers will more accurately predict PD-1/PD-L1 therapy response and identify patients who require combination strategies targeting tumor metabolism and/or DNA repair–associated pathways to restore effective antitumor immunity.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PD-L1, bTMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PD-L1: PD-L1 expression nevertheless fail to respond to PD-1/PD-L1 therapy; why does apparent immune engagement not translate to effective antitumor immunity? 2) Tier 1 paths used: (a) PD-L1 ↔ bTMB (core path 1) — suggests neoantigen-driven adaptive PD-L1 upregulation; (b) PD-L1 ↔ CD4/CD8 TILs (core path 3) — links checkpoint expression to tissue T cell presence.

bTMB: bTMB (core path 1) — suggests neoantigen-driven adaptive PD-L1 upregulation; (b) PD-L1 ↔ CD4/CD8 TILs (core path 3) — links checkpoint expression to tissue T cell presence.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	LRPPRC	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	SARIFA-positivity	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	TG	Supporting modifier from knowledge graph
Secondary	SHBG	Supporting modifier from knowledge graph
Secondary	IGF-1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- LRPPRC: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.